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Programme: F.E. (All Branches)

Sample paper Internal Assessment Test II (A.Y. 2026-27)

Subject: Applied Physics **NEP–2020** **Semester: I**

Max. Marks: 15 **Date: 23/11/2026** **Time: 3:00 to 4:00**

Note: All questions are compulsory.

Q. No.	Questions	Marks	BL	CO
Q.1	**Attempt any two of the following:**			
1(a)	Illustrate the energy band structures of direct and indirect semiconductors with proper labeling and explain the key differences between them.	3	L2	CO6
1(b)	(i) If electron concentration $n = 5.0 \times 10^{22}\,\mathrm{m}^{-3}$ with $\mu_n = 0.12\,\mathrm{m}^2/\mathrm{V}\cdot\mathrm{s}$; hole concentration $p = 1.0 \times 10^{22}\,\mathrm{m}^{-3}$ with $\mu_p = 0.05\,\mathrm{m}^2/\mathrm{V}\cdot\mathrm{s}$, calculate the total conductivity σ. (ii) A metal has Fermi energy $E_F = 6.0\,\mathrm{eV}$. Compute k_BT in eV at $T = 500\,\mathrm{K}$ and estimate the occupation probability at $E = E_F + k_BT$.	3	L3	CO6
1(c)	Formulate the expression for the intrinsic Fermi energy (E_i) and illustrate its position between the conduction band (E_c) and valence band (E_v) with a neat energy band diagram.	3	L3	CO6
Q.2	**Attempt any one of the following:**			
2(a)	Given the vector field $\vec{A} = 3x^2\hat{i} + 4xy^2\hat{j} + yz\hat{k}$, calculate: (a) Divergence of $\vec{A}$ (b) Curl of $\vec{A}$	4	L3	CO4
2(b)	Obtain Maxwell's equations from the basic laws of electricity and magnetism and explain the physical implications of these equations for electromagnetic field theory.	4	L3	CO4
Q.3	**Attempt any one of the following:**			
3(a)	Discuss de Broglie's hypothesis and the properties of matter waves. Evaluate the wavelength of an electron: • Accelerated through a potential difference of 150 V • In thermal equilibrium at 400 K • Having kinetic energy of 20 eV	5	L2, L3	CO5
3(b)	Derive the time-dependent and time-independent Schrödinger wave equations.	5	L2, L3	CO5

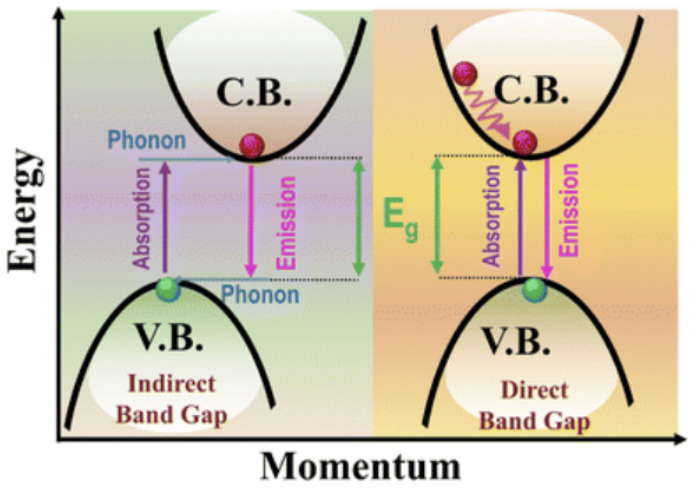
BL: Bloom's Taxonomy Levels
(L1: Remembering, L2: Understanding, L3: Applying, L4: Analyzing, L5: Evaluating, L6: Creating)
CO: Course Outcomes

Model Solutions & Marking Scheme

Question 1(a) **Total: 3 Marks**

Illustrate the energy band structures of direct and indirect semiconductors with proper labeling and explain the key differences between them.

Energy Band Structure Diagram: [1 M]



Key Differences: [2 M]

Feature	Direct Bandgap Semiconductor	Indirect Bandgap Semiconductor
Band Extrema	Conduction band minimum and valence band maximum occur at the same momentum ($k = 0$).	Conduction band minimum and valence band maximum occur at different k-values.
Electron Transition	Transition occurs directly by photon emission/absorption without momentum change.	Transition requires a simultaneous change in momentum, assisted by a phonon.
Recombination Process	Predominantly radiative; high photon emission efficiency.	Predominantly non-radiative; emitted energy is dissipated as heat/vibrations.
Absorption Coefficient	High; sharp absorption edge.	Relatively low near the band edge.
Common Examples	Gallium Arsenide (GaAs), Indium Phosphide (InP).	Silicon (Si), Germanium (Ge).
Typical Applications	Optoelectronic devices: LEDs, laser diodes, photodetectors.	Electronic devices: rectifiers, transistors, solar photovoltaic cells.

Question 1(b) **Total: 3 Marks**

Conductivity calculation and Fermi–Dirac distribution occupation probability.

Part 1: Calculation of Electrical Conductivity (σ) [1.5 M]

- Electron density: $n = 5.0 \times 10^{22}\,\mathrm{m}^{-3}$, Mobility: $\mu_n = 0.12\,\mathrm{m}^2/\mathrm{V}\cdot\mathrm{s}$
- Hole density: $p = 1.0 \times 10^{22}\,\mathrm{m}^{-3}$, Mobility: $\mu_p = 0.05\,\mathrm{m}^2/\mathrm{V}\cdot\mathrm{s}$
- Elementary charge: $e = 1.602 \times 10^{-19}\,\mathrm{C}$

Formula for total conductivity:

$$\sigma = e(n\mu_n + p\mu_p)$$

Evaluating individual terms:

$$\begin{aligned} n\mu_n &= (5.0 \times 10^{22})(0.12) = 6.0 \times 10^{21}\,\mathrm{m^{-1}V^{-1}s^{-1}} \\ p\mu_p &= (1.0 \times 10^{22})(0.05) = 0.5 \times 10^{21}\,\mathrm{m^{-1}V^{-1}s^{-1}} \\ n\mu_n + p\mu_p &= 6.5 \times 10^{21}\,\mathrm{m^{-1}V^{-1}s^{-1}} \end{aligned}$$

Substituting into conductivity:

$$\sigma = (1.602 \times 10^{-19}\,\mathrm{C}) \times (6.5 \times 10^{21}) = \mathbf{1.041 \times 10^{3}}\,\mathrm{S/m}$$

Part 2: Fermi–Dirac Distribution Probability **[1.5 M]**

Given $E_F = 6.0\,\mathrm{eV}$ and $T = 500\,\mathrm{K}$. The thermal energy is:

$$k_B T = \frac{(1.3806 \times 10^{-23}\,\mathrm{J/K})(500\,\mathrm{K})}{1.602 \times 10^{-19}\,\mathrm{J/eV}} \approx 0.0431\,\mathrm{eV}$$

The Fermi–Dirac occupation probability function is:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

For $E = E_F + k_B T$, the exponent simplifies to $\frac{E-E_F}{k_B T} = 1$:

$$f(E) = \frac{1}{1 + e^{1}} = \frac{1}{1 + 2.7183} = \frac{1}{3.7183} \approx \mathbf{0.269} \quad (26.9\%)$$

Question 1(c) **Total: 3 Marks**

Derivation of intrinsic Fermi level (E_i) and energy band representation.

Mathematical Derivation: **[2 M]**

In an intrinsic semiconductor at thermal equilibrium, electron concentration n equals hole concentration p ($n = p$). Hence, the probability of an electron occupying the conduction band equals the probability of finding a hole in the valence band:

$$f(E_c) + f(E_v) = 1 \tag{1}$$

From the Fermi–Dirac distribution:

$$f(E_c) = \frac{1}{1 + e^{(E_c - E_F)/kT}}, \qquad f(E_v) = \frac{1}{1 + e^{(E_F - E_v)/kT}}$$

Substituting into Eq. (1):

$$\frac{1}{1 + e^{(E_c - E_F)/kT}} + \frac{1}{1 + e^{(E_F - E_v)/kT}} = 1$$

Combining fractions over a common denominator:

$$\frac{2 + e^{(E_F - E_v)/kT} + e^{(E_c - E_F)/kT}}{\left[1 + e^{(E_c - E_F)/kT}\right]\left[1 + e^{(E_F - E_v)/kT}\right]} = 1$$

Expanding the denominator on the right-hand side:

$$2 + e^{(E_F - E_v)/kT} + e^{(E_c - E_F)/kT} = 1 + e^{(E_c - E_F)/kT} + e^{(E_F - E_v)/kT} + e^{(E_c + E_v - 2E_F)/kT}$$

Canceling identical terms from both sides gives:

$$e^{(E_c + E_v - 2E_F)/kT} = 1$$

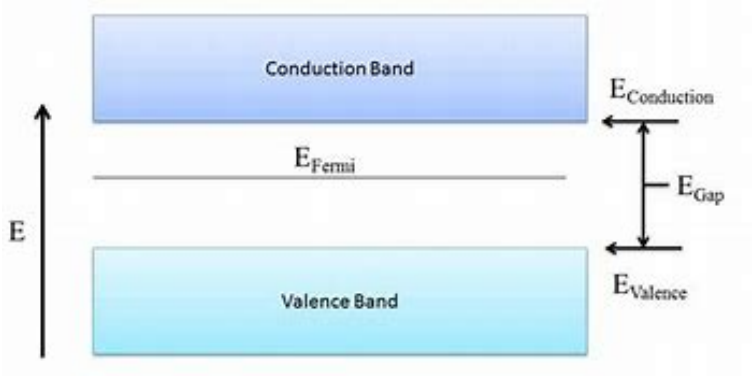
Taking the natural logarithm on both sides:

$$\frac{E_c + E_v - 2E_F}{kT} = \ln(1) = 0 \implies E_c + E_v - 2E_F = 0$$

$$\mathbf{E_F = \frac{E_c + E_v}{2}}$$

Conclusion: At $T = 0\,\text{K}$, the intrinsic Fermi level lies exactly midway between the conduction and valence band edges.

Energy Band Diagram: **[1 M]**



Question 2(a) **Total: 4 Marks**

Calculate the divergence and curl of vector field $\vec{A} = 3x^2\hat{i} + 4xy^2\hat{j} + yz\hat{k}$.

Given vector field: $\vec{A} = P\hat{i} + Q\hat{j} + R\hat{k} = 3x^2\hat{i} + 4xy^2\hat{j} + yz\hat{k}$ where $P = 3x^2$, $Q = 4xy^2$, $R = yz$.

(a) Divergence of $\vec{A}$: **[2 M]**

$$\nabla \cdot \vec{A} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Calculating the partial derivatives:

$$\frac{\partial}{\partial x}(3x^2) = 6x, \qquad \frac{\partial}{\partial y}(4xy^2) = 8xy, \qquad \frac{\partial}{\partial z}(yz) = y$$

$$\boldsymbol{\nabla \cdot \vec{A} = 6x + 8xy + y}$$

(b) Curl of $\vec{A}$: **[2 M]**

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 & 4xy^2 & yz \end{vmatrix} = \hat{i}\left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) - \hat{j}\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + \hat{k}\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$$

Evaluating the components:

$$\begin{aligned} \hat{i}\text{-component:} &\quad \frac{\partial}{\partial y}(yz) - \frac{\partial}{\partial z}(4xy^2) = z - 0 = z \\ \hat{j}\text{-component:} &\quad \frac{\partial}{\partial x}(yz) - \frac{\partial}{\partial z}(3x^2) = 0 - 0 = 0 \\ \hat{k}\text{-component:} &\quad \frac{\partial}{\partial x}(4xy^2) - \frac{\partial}{\partial y}(3x^2) = 4y^2 - 0 = 4y^2 \end{aligned}$$

$$\boldsymbol{\nabla \times \vec{A} = z\hat{i} + 4y^2\hat{k}}$$

Question 2(b) **Total: 4 Marks**

Maxwell's equations: Derivation from fundamental laws and physical implications.

Summary of Maxwell's Equations and Formulations: **[3 M]**

Law	Integral Form	Differential	Physical Significance
Gauss's Law (Electrostatics)	$\oint_S \vec{D} \cdot d\vec{S} = \int_V \rho_v \, dv$	$\nabla \cdot \vec{D} = \rho_v$	Electric field lines diverge from positive charge sources and converge at sinks; static charges are sources of $\vec{D}$.
Gauss's Law (Magnetostatics)	$\oint_S \vec{B} \cdot d\vec{S} = 0$	$\nabla \cdot \vec{B} = 0$	Magnetic flux lines are closed continuous loops. Isolated magnetic poles (monopoles) do not exist in nature.
Faraday's Law of Induction	$\oint_C \vec{E} \cdot d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	A time-varying magnetic flux induces a non-electrostatic, circulation-type electric field (basis of generators and transformers).
Ampère–Maxwell Law	$\oint_C \vec{H} \cdot d\vec{l} = \int_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t}\right) \cdot d\vec{S}$	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	Magnetic fields originate both from conduction current $\vec{J}$ and time-varying electric displacement current $\frac{\partial \vec{D}}{\partial t}$.

Implications for Electromagnetic Field Theory: **[1 M]**

Together, Faraday's Law and the Ampère–Maxwell Law predict self-propagating transverse electromagnetic waves traveling at speed $c = 1/\sqrt{\mu_0 \varepsilon_0}$ in vacuum, uniting optics, electricity, and magnetism into a single comprehensive framework.

Question 3(a) **Total: 5 Marks**

De Broglie's hypothesis, matter wave characteristics, and numerical wavelength evaluations.

1. De Broglie Hypothesis: **[1 M]**

In 1924, Louis de Broglie hypothesized that dual nature is a general characteristic of the universe. Any moving matter particle of mass m and velocity v has an associated matter wave with wavelength:

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where $h = 6.626 \times 10^{-34}\,\mathrm{J \cdot s}$ is Planck's constant.

2. Key Properties of Matter Waves: **[1 M]**

- **Inverse Momentum Dependency:** Wavelength is inversely proportional to momentum ($\lambda \propto 1/p$); lighter particles have longer, more measurable matter wavelengths.
- **Non-Electromagnetic:** Matter waves describe probability amplitude of finding a particle; they are neither electric nor magnetic in nature.
- **Velocities:** Phase velocity ($v_p = \frac{\omega}{k} = \frac{c^2}{v} > c$) exceeds c, but group velocity ($v_g = \frac{d\omega}{dk} = v$) corresponds directly to the physical particle speed.

3. Numerical Evaluations of Electron Wavelength (λ):

(a) **Accelerated through potential $V = 150\,\mathbf{V}$:** **[1 M]**

$$\lambda = \frac{h}{\sqrt{2 m_e e V}} = \frac{1.227}{\sqrt{V}}\,\mathrm{nm} = \frac{1.227}{\sqrt{150}}\,\mathrm{nm} = \frac{1.227}{12.247}\,\mathrm{nm} = \mathbf{0.1002}\,\mathrm{nm} \quad (1.002\,\text{Å})$$

(b) **Thermal equilibrium at $T = 400\,\mathbf{K}$ (average translational KE $= \frac{3}{2}k_BT$):** **[1 M]**

$$\lambda = \frac{h}{\sqrt{3m_e k_B T}} = \frac{6.626 \times 10^{-34}}{\sqrt{3 \times (9.109 \times 10^{-31}) \times (1.381 \times 10^{-23}) \times 400}} = \mathbf{0.5436}\,\text{nm}$$

(c) **Kinetic Energy $E = 20\,\mathbf{eV}$ ($= 20 \times 1.602 \times 10^{-19}\,\mathbf{J} = 3.204 \times 10^{-18}\,\mathbf{J}$):** **[1 M]**

$$\lambda = \frac{h}{\sqrt{2m_e E}} = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times (9.109 \times 10^{-31}) \times (3.204 \times 10^{-18})}} = \mathbf{0.2745}\,\text{nm} \quad (2.745\,\text{Å})$$

Question 3(b) **Total: 5 Marks**

Derivation of Schrödinger time-dependent and time-independent wave equations.

Part I: Schrödinger Time-Dependent Wave Equation (STWE) **[2.5 M]**

A matter wave associated with a free particle moving along the x-direction is described by:

$$\psi(x,t) = \psi_0 e^{i(kx-\omega t)} \tag{1}$$

From de Broglie's and Planck's relations:

$$k = \frac{2\pi}{\lambda} = \frac{p}{\hbar}, \qquad \omega = 2\pi\nu = \frac{E}{\hbar} \quad \left(\text{where } \hbar = \frac{h}{2\pi}\right)$$

Substituting into Eq. (1):

$$\psi(x,t) = \psi_0 \exp\left[\frac{i}{\hbar}(px - Et)\right] \tag{2}$$

Differentiating Eq. (2) with respect to time t:

$$\frac{\partial\psi}{\partial t} = -\frac{iE}{\hbar}\psi \implies E\psi = i\hbar\frac{\partial\psi}{\partial t} \tag{3}$$

Differentiating Eq. (2) twice with respect to space coordinate x:

$$\frac{\partial\psi}{\partial x} = \frac{ip}{\hbar}\psi \implies \frac{\partial^2\psi}{\partial x^2} = -\frac{p^2}{\hbar^2}\psi \implies p^2\psi = -\hbar^2\frac{\partial^2\psi}{\partial x^2} \tag{4}$$

The classical Hamiltonian (total energy) is $E = \frac{p^2}{2m} + V(x)$. Operating on $\psi(x,t)$:

$$E\psi = \frac{p^2}{2m}\psi + V\psi$$

Substituting the operator results from Eq. (3) and Eq. (4):

$$\boldsymbol{i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi(x,t)}$$

This represents the 1D **Schrödinger Time-Dependent Equation**.

Part II: Schrödinger Time-Independent Wave Equation (STIWE) **[2.5 M]**

Assuming stationary states, separate variables into spatial and temporal parts:

$$\psi(x,t) = \psi(x)\phi(t) = \psi(x)e^{-i\omega t} = \psi(x)e^{-iEt/\hbar}$$

Evaluating the required derivatives:

$$\frac{\partial\psi(x,t)}{\partial t} = -\frac{iE}{\hbar}\psi(x)e^{-iEt/\hbar}, \qquad \frac{\partial^2\psi(x,t)}{\partial x^2} = \frac{d^2\psi(x)}{dx^2}e^{-iEt/\hbar}$$

Substituting these into the time-dependent wave equation:

$$i\hbar\left(-\frac{iE}{\hbar}\psi(x)e^{-iEt/\hbar}\right) = -\frac{\hbar^2}{2m}\left(\frac{d^2\psi(x)}{dx^2}e^{-iEt/\hbar}\right) + V\psi(x)e^{-iEt/\hbar}$$

Since $-i^2 = 1$ and the factor $e^{-iEt/\hbar} \neq 0$, divide throughout by $e^{-iEt/\hbar}$:

$$E\psi(x) = -\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V\psi(x)$$

Rearranging:

$$(E-V)\psi(x) = -\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} \implies \frac{d^2\psi(x)}{dx^2} = -\frac{2m}{\hbar^2}(E-V)\psi(x)$$

$$\boldsymbol{\frac{d^2\psi(x)}{dx^2} + \frac{2m}{\hbar^2}(E-V)\psi(x) = 0}$$

This is the 1D **Schrödinger Time-Independent Equation**.